

On farm productivity is measured by the ratio of outputs produced (such as crops, livestock and wool) to the inputs used (such as land, labour, materials and services).

Improving your productivity means you produce the same amount of outputs using fewer inputs, or more outputs with the same inputs.

There are actions you can do now to improve your farm productivity and lower the emissions intensity of your production. Stacking small productivity improvements can have an impact on your emissions profile.

What you can do

To improve your productivity, you could explore ways to:

- improve your nitrogen use efficiency (NUE)
- manage herd and pasture
- adopt new technology.

Improve your nitrogen use efficiency

For cropping or pasture systems, consider if you can find ways to improve your NUE.

Nitrogen is essential for plant growth. When there's more nitrogen in the soil, crops and pasture usually grow better. Farmers often add nitrogen fertilisers to boost this growth.

But an estimated **40-60 % of nitrogen inputs** into grazing and cropping systems are lost to the environment (University of Melbourne). This is a significant cost to the farmer and the environment.

Improving your NUE means increasing the proportion of applied nitrogen that is captured in products and reducing nitrous oxide losses. You can do this by applying the 4 Rs:

- right product
- right rate
- right time
- right place.

NUE can reduce nitrogen losses, improve productivity, lower your input costs and reduce nitrous oxide emissions. It can also benefit the environment by reducing nitrogen leaching and runoff which pollutes waterways and groundwater.

Learn more about NUE:

- [Nitrogen fertilisers — improving efficiency and saving money](#) (Agriculture Victoria)
- [Practice Guide: Improving Nitrogen Cycling and Efficiency](#) (Soils for Life)

Manage herd and pasture

For livestock operations, you may consider if there are ways to improve the efficiency and productivity of your herd. These strategies can often reduce emissions intensity.

Strategies you could consider include:

- enhancing nutrition by planting improved pastures and introducing phosphorous supplements
- installing new fences to enable improved management of the joining period
- increasing density of water points
- selecting animals for breeding with increased reproductive rates and shorter finishing times
- culling unproductive breeding stock
- minimising young stock losses through good husbandry and provision of adequate shelter.

This is not an exhaustive list. See more guidance on herd and pasture management practices:

- [Livestock methane and nitrogen emissions](#) (Agriculture Victoria)
- [Managing livestock to reduce emissions](#) (NSW DPI)

You can also visit Meat & Livestock Australia for:

- [Resource hubs](#): resources on beef, sheep, goats and grazing land management
- [Carbon EDGE training program](#): training on farm activities that reduce emissions while improving productivity.

Adopt new technology

Adoption of new technology can help to minimise inputs, such as fertiliser use. It can also help to maximise crop yields or livestock growth, fertility and survival. New technologies could include:

- precision agriculture, which uses technology to find variations in different areas of land and use this information to tailor management responses
- high yield crop varieties
- genetic tools that provide breeding values for a range of livestock traits.

These technologies can also help to improve your [energy efficiency](#), and reduce the [emissions intensity](#) of your production.

To keep up to date with emerging technologies that improve farm productivity, you can explore upcoming events hosted by your relevant [Research and Development Corporation](#) (DAFF).

Find out more about [precision agriculture](#) (AgriFutures).

Resources to help you improve productivity

Explore further key resources on improving productivity.

See more

Learn more about how to improve productivity:

- [ABARES](#): Learn more about farm productivity.
- [Research and Development Corporations \(RDC\)](#): Explore tools, research and events from your relevant RDC.
- [CEFC](#): Learn more about emissions reduction activities that can improve productivity using the Towards Net Zero Agriculture Pathfinder.

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/climatechange/carbon-farming-outreach-program/hub/ways-reduce-emissions/improve-productivity>